UTILS

Utils contain logging functions and the embedding and API call functions.
In the API call function generate_text() there is the **temperature hyperparameter** which is potentially tunable.
Temperature could be either co-evolved with the prompts, so added to the single node (which would allow an initialization with 50 different individuals instead of initializing a population of 50 equal individuals), but it would require another ad-hoc mutation. The other possibility is to keep it fixed at some value, and assume that the success of the experiment is determined by the evolution of the prompts themselves more than on the settings like the temperature.
We will keep 0.0 for evaluation as baseline to minimize non-determinism and at 0.8 in mutations as empirical value to ensure variability

TRAITS

Trait is a class that encapsulates the individual node into a runnable entity that gets previous context as input. It combines context with the node's internal instructions and calls a function that performs an API call to the LLM server to get a response. It outputs an estimate of the total amount of input tokens, output tokens, API call processing time and the final answer.
I perfected the API to directly return the precise count of in and out tokens and the TTFT and ITL - tokens per second.
*Here a possible modification is to how the **previous context and the next node's instructions are joined**. A standard way would be to compose a history in a chat-like way, so with messages exchanged between a user and the model. Custom methods are possible but would require manual prompt engineering which is based on experience and subjective heuristics so it would be harder to justify in a possible paper explanation.*
We implement it as a chat to not lose intermediate prompts and not disrupt internal coherence of the reasoning process.

PHENOTYPE

Phenotype is a class that takes as input a single problem from the dataset and makes an individual execute it. It gets the execution order of the nodes from the genome and iteratively calls the traits on the single nodes and combines the parents' responses to form the context. It also accumulates stats for time, tokens and number of steps.
*Here the only possible modification is to the **context formation process**. The critical step is to define the system instructions, which fall outside the user-model conversation. These could be set either manually or justified with a review from a SOTA AI model.*
We defined the system instructions manually with help from a SOTA model to validate and perfect them.

GENE

Genes are formed using the class PromptNode, which has id (unique), innovation number, instructions (prompt), name (basically a very short summary of the scope of the instructions) and embedding (of the instruction, to semantically compare nodes).
Genes are assembled using the Connection class, which is composed of an in_node, an out_node, an 'enabled' flag and an innovation number derived from the hash of input node innovation number and output node innovation number. An innovation number registry is kept to keep track of the semantics associated with all the innovation numbers and it ensures that whenever a new gene is created or an old one is mutated, it's routed toward the

right innovation number. This registry groups prompts by highest average semantic similarity using a threshold empirically set according to human perception preferences.

GENOME

Genome is a class that contains all the logic necessary to combine Genes and Connections. It handles the addition and removal of nodes, the logic to get the execution order and the checks to ensure the acyclicity of the graph using Khan's algorithm.

MUTATIONS

Mutations are managed by the mutator class that performs, both for architectural mutations (remove and add nodes and connections) and for gene-level mutations (expand, simplify, reformulate), plus a mutation that acts in both levels at the same time (split node)
It is configured to allow dynamic probabilities for all mutations, first discriminating between architectural and gene-level, then discriminating between subtypes:

```
   DEFAULT_CONFIG = {
       "p_architectural_event": 0.30, # architectural mutation change
       "p_mutate_node": 0.10,         # gene-level mutation chance

       # Relative weights for Architectural mutations
       "arch_probs": {
           MutType.ARCH_ADD_NODE: 0.25,
           MutType.ARCH_REMOVE_NODE: 0.25,
           MutType.ARCH_ADD_CONN: 0.25,
           MutType.ARCH_REMOVE_CONN: 0.25,
       },

       # Relative weights for Gene mutations
       "gene_probs": {
           MutType.GENE_EXPAND: 0.25,
           MutType.GENE_SIMPLIFY: 0.25,
           MutType.GENE_REFORMULATE: 0.30,
           MutType.GENE_SPLIT: 0.20  # 20% chance to split if selected
       }
   }
```

these probabilities are dynamically modified using the following approach:

```
   def get_dynamic_config(generation: int, node_count: int,
connections_count: int) -> dict:
       """
       Calculates mutation probabilities using a strict mathematical
'Phase Shift' strategy.
       Architectural mutations act as a self-correcting thermostat
centered at N=4 nodes,
       with a hard cap at N=6.
```

```
        Gene mutations shift from early Cognitive Structuring to late
Linguistic Compression.
        """

        # --- 1. Global Rate Calculations ---
        # Architectural Decay: Cools down global topology changes over
generations
        # Architectural Decay: Cools down global topology changes
smoothly
        if generation < 2:
            p_arch = 0.0
            p_gene = 0.8
        else:
            # Shift the exponent so Gen 2 = 0
            decay_steps = generation - 2

            # Starts and decays by 15% per generation
            p_arch = max(0.4, 0.70 * (0.85 ** decay_steps))

            # Starts and decays by 15% per generation
            p_gene = max(0.3, 0.70 * (0.85 ** decay_steps))

        # --- 2. Architectural Probabilities ---
        max_C = (node_count * (node_count - 1)) / 2.0
        min_C = node_count -1
        remove_conn = (connections_count - min_C) / (max_C - min_C)

        add_node = 0.0
        if node_count == 2:
            add_node = 0.7
        elif node_count == 3:
            add_node = 0.6
        elif node_count == 4:
            add_node = 0.5
        elif node_count >= 5:
            add_node = 0.0
        remove_node = 1.0 - add_node
        add_conn = 1.0 - remove_conn

        if node_count <= 1:
            p_add_node = 0.0
            p_remove_node = 0.0
            p_add_conn = add_conn
```

```
            p_remove_conn = remove_conn
        else:
            p_add_node = add_node * 0.65
            p_remove_node = remove_node * 0.65
            p_add_conn = add_conn * 0.35
            p_remove_conn = remove_conn * 0.35

        # --- 3. Gene Probabilities (Structuring -> Compression) ---
        split = max(0.0, 0.5 - node_count * 0.1)
        others = (1.0 - split) / 5

        return {
            "p_architectural_event": p_arch,
            "p_mutate_node": p_gene,

            "arch_probs": {
                MutType.ARCH_ADD_NODE:    p_add_node,
                MutType.ARCH_REMOVE_NODE: p_remove_node,
                MutType.ARCH_ADD_CONN:    p_add_conn,
                MutType.ARCH_REMOVE_CONN: p_remove_conn,
            },

            "gene_probs": {
                MutType.GENE_SPLIT:            split,
                MutType.GENE_INJECT_REASONING: others,
                MutType.GENE_EXPAND:           others,
                MutType.GENE_ADD_PERSONA:      others,
                MutType.GENE_REFORMULATE:      others,
                MutType.GENE_SIMPLIFY:         others
            }
        }
```

Mutation probabilities are set dynamically to adjust to the topology of the individual. They ensure topology cannot be broken (maintain DAG structure), and that all valid topologies can be explored.

The prompts are defined manually with the help of a SOTA model to review and perfect them, as for the system instructions.

Maturity threshold (target node count) was introduced to account for attention dilution in small models and is backed by separate ad-hoc testing.

1. Architectural Probabilities (arch_probs) Logic

To meet your exact constraints, we define two mathematical balances, while proportion between the two was set empirically.

Balance A: Node Count (Target N=4)

- *We use a linear decay with a hard cap as suggested by our separate testing*

Balance B: Connection Density (The DAG Structural Integrity)

- *In a Directed Acyclic Graph, the absolute maximum number of possible connections for N nodes is C_max = N(N-1)/2*
- *In a Directed Acyclic Graph, the absolute minimum number of possible connections for N nodes is C_min = N-1*
- *Since the DAG has 1 in node and 1 out node, we derived a formula that varies from 0 when the DAG has minimal connectivity to 1 when the DAG has maximal connectivity and used it as "remove connection" probability.*

2. Gene-Level Probabilities (gene_probs) Logic

Because split mutation is a mutation both at gene level and at architecture level, we link it to the nodes count to ensure the hard cap on complexity is respected. The other mutations are equally probable.

CROSSOVER
Crossover is implemented in the Crossover class, which originally did not correctly reflect NEAT principles, as it blindly inherited all non-matching genes, without caring about fitness. I reimplemented it to adhere to NEAT logic. Adding a new function to check and clean eventual mismatches to ensure the validity of the offspring. I also added a set of tests to ensure the code is working properly
Probabilities of inheriting a common node or connection from either parents at 50/50 is the NEAT standard, there is nothing to change here.
Like NEAT, 25% of individuals are generated only via mutation without crossover.
To avoid graph consistency issues when two parents have equal performance, a random one is chosen as the better parent.

SELECTION
The choice of the selection strategy was determined by the following reasoning process:

1. As demonstrated by the crossover unit tests, merging graph topologies is highly destructive. Even with topological pruning, a child of two brilliant parents might end up with a fundamentally broken logic flow. If we do not use **elitism**, we risk losing our best architectures to bad dice rolls in the mutation/crossover phase, causing the engine's performance to violently regress. That's why we always keep the 3 best individuals in a generation, plus we also implement NEAT-inspired elitism where we also keep 1 elite for every 5 members of a species (NEAT just keeps 1 if the species has at least 5 individuals).
2. Prompt-based fitness scores can be highly non-linear or difficult to normalize cleanly. Tournament selection doesn't care about the exact mathematical difference between scores; it only cares about rank (Individual A > Individual B). If one graph accidentally

discovers an exploit that gives it a massive fitness score, proportional selection will cause it to instantly dominate the breeding pool, killing diversity. **Rank based selection** inherently limits how many times the top genome can be selected. This deviation from standard NEAT is necessary due to the fundamentally different natures of the individuals.

3. A Plus strategy ensures that a parent is only replaced if a child genuinely outperforms it. For complex graph evolution, this assumption is false; offspring are often worse, so a plus strategy would destroy innovation, that's why we will stick with standard NEAT **Comma strategy**.

This combination guarantees that your best logic chains are never lost (Elitism), prevents super-individuals from wiping out the population (RankBased), and protects new, fragile architectures while they optimize (Comma Strategy).

Selection is implemented in the Selection submodule (which contains implementations that allow multiple combinations for the selection strategy), with additional pieces in the Evolution Manager and in the Generation Manager.
The code for this messy section was cleaned and reorganized into Selection and SpeciesBreeder, removing all unnecessary functions. On top of this, all the hyperparameters were identified and extracted as variables to allow easy tuning.
Selection pressure *(selection.py) is set at 1.5 as standard and* ***elitism ratio*** *(species_breeder.py) is at 0.2 to maintain the NEAT idea to not have elitism for species smaller than 5 individuals, but we get more than 1 elite individual for bigger species, compared to NEAT which keeps 1 max, since our crossover and mutations are potentially more harmful. These hyperparameters are not that influential and are thus set empirically based on standards and on suggestions from SOTA models.*

SPECIES
Species were previously implemented in the Species class, with parts in the Evolution Manager and in the Generation Manager, with complex speciation mechanisms and a fixed threshold. I simplified the logic to better resemble the NEAT proposal, and implemented a dynamic threshold to keep a fixed number of species in each generation, and thus avoid both divergence into many small species and convergence into a single one.

INPUT: A flat list of evaluated `AgentGenome` objects.

1. **Routing (Speciation):** The `SpeciationEngine` iterates through the genomes. It calculates the topological and semantic distance (`compatibility_distance`) between each genome and the existing species' Representatives. If the distance is below the `compatibility_threshold`, the genome joins that species. If not, it creates a new species. A primordial species is kept to put all 1-node individuals that do not belong to other species. This is necessary as 1-node individuals would otherwise generate an explosion in the number of species since compatibility between them is all or nothing.

   Compatibility is computed as len(shared_genes) / max_nodes . This choice is due to empirical findings.

2. **The Thermostat (Dynamic Thresholding):** The engine checks the total number of species created. If it's above or below an ideal number (4, set empirically to ensure every species has enough individuals to perform a meaningful breeding), it adjusts the threshold by an amount set empirically to avoid strong oscillations to steer the next generation toward the ideal number. It has to be noted that the system automatically deviates toward 6-7 species.
3. **Stagnation Culling & Failsafe:** The engine checks if any species has failed to beat its historical high score in the last X generations. If so, it is marked for death. *Failsafe:* If this mass-culling would drop the total active species below 2, the engine spares the historically best dying species and resets its stagnation counter NEAT-based approach.
4. **Resource Allocation (Fitness Sharing):** The engine calculates the average fitness of each surviving species. It sums these averages and allocates the 50 next-generation slots proportionally using fitness sharing like NEAT.
5. **Breeding (The Micro-Layer):** The engine passes the members and the allocated `target_size` to the `SpeciesBreeder`. The breeder calculates how many elites to clone, uses `RankBasedSelection` to pick parents, and generates the remaining children via crossover and mutation. It resets the children's fitness to `0.0`.
6. **State Reset:** Surviving species pick a new random Representative from their current members and clear their member lists to prepare for the next generation.

OUTPUT: The list of `AgentGenome` objects for the new generation.

FITNESS

Fitness was implemented as a loss using a multi-parameter approach similar to original NEAT, with complex cosine-similarity operations and multiple weight-hyperparameters.
I refactored the code to simplify the fitness computation as a score, so now higher is better, and it never goes negative, so as not to break assignments to the species. This new score system is given by the number of correct answers minus a penalty given by the token usage, which is directly proportional to the complexity of the graph, so we do not need another penalty for that. The weight of the token count is dynamically adjusted to reflect the previous generation, so that if more complex individuals effectively outperform smaller ones, and even bigger ones would work even better, the penalty is gradually adjusted to move towards the best solution possible. I also added an adjustable threshold that if an individual misses too many answers in a row it’s considered faulty and the evaluation stops prematurely to save resources and time.
The weight of the tokens is a hyperparameter that has to be manually set as it represents a user preference.

MAIN / UTILS

I re-wrote the main orchestration functions to correctly work with the fixed codebase and perfected the utils functions to reflect the changes as well.

LLM server

I reimplemented the vllm server to support the qwen 1.7B model and modified the codebase to batch queries so to run the up to 50 individuals at the same time leveraging vllm parallelization capabilities.

I also implemented a new ollama server to run LFM 2.5 1.2B models for further testing and adapted the code to natively handle both servers
It has to be noted that although LFM models are faster for single request, vllm handles concurrency incredibly better than ollama, so from my tests I saw that with vllm we can easily surpass 1-2 k TPS, while ollama is capped around 220 TPS, 5 to 10 times slower than VLLM for concurrent tasks. For single use, ollama easily stays over 120 TPS, while vllm is capped between 50 and 80 TPS.
Document improvement in performance between single instance and parallelization.

DEBUGGING
Debugging was done to ensure the system runs properly. I ensured it worked by launching a 6 generations run, in which fitness slightly increased and also complexity started to rise. It's notable how using Gemma 3 with vLLM batching allowed the first 1 node generations to run for 50 individuals with batches of 50 problems per individual at a blazing speed of around 10 seconds total, while doubling the complexity rises the time at around 40 seconds per generation, which is still incredibly fast. I nodded that the number of species stayed balanced and there was no explosion nor collapse, but the plotter still has some issues in representing the different species. I also noted that dynamic parameters for token usage and speciation seem to work fine.

We can think of sub-problems or simplified versions of the problem to test the hyperparameter tuning in a safer and easier space.

BASELINES
Baselines were implemented to resemble standard practices like zero shot with heavy prompt engineering and few shots. Both baselines were refined through iterative testing and consulting with a SOTA model.

Level 02 Hops: 5101
Level 03 Hops: 5103
Level 04 Hops: 5101
Level 05 Hops: 185
Level 06 Hops: 105
Level 07 Hops: 155
Level 08 Hops: 135
Level 09 Hops: 124
Level 10 Hops: 122
==
🎯 STRATIFIED BASELINE REPORT t=0
==
Execution Time: 125.51 seconds
Execution Time: 143.84 seconds with t=0.01
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 12.92% (659/5101)
Level 03 Hops: Accuracy 16.03% (818/5103)
Level 04 Hops: Accuracy 13.86% (707/5101)
Level 05 Hops: Accuracy 11.35% (21/185)
Level 06 Hops: Accuracy 04.76% (5/105)

Level 07 Hops: Accuracy 03.23% (5/155)
Level 08 Hops: Accuracy 08.15% (11/135)
Level 09 Hops: Accuracy 06.45% (8/124)
Level 10 Hops: Accuracy 02.46% (3/122)

Overall Dataset Accuracy: 13.87%
Overall Average Fitness: 0.1241

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16131 (100.0%)
Exactly 2-Word Answers: 0 (0.0%)
==========
==========
🎯 STRATIFIED BASELINE REPORT (t=0.5)
==========
Execution Time: 145.34 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 13.17% (672/5101)
Level 03 Hops: Accuracy 16.19% (826/5103)
Level 04 Hops: Accuracy 13.82% (705/5101)
Level 05 Hops: Accuracy 11.89% (22/185)
Level 06 Hops: Accuracy 03.81% (4/105)
Level 07 Hops: Accuracy 03.23% (5/155)
Level 08 Hops: Accuracy 08.89% (12/135)
Level 09 Hops: Accuracy 06.45% (8/124)
Level 10 Hops: Accuracy 02.46% (3/122)

Overall Dataset Accuracy: 13.99%
Overall Average Fitness: 0.1252

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16130 (100.0%)
Exactly 2-Word Answers: 1 (0.0%)
==========
==========
🎯 STRATIFIED BASELINE REPORT (t=1)
==========
Execution Time: 145.90 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 12.86% (656/5101)
Level 03 Hops: Accuracy 15.89% (811/5103)
Level 04 Hops: Accuracy 13.66% (697/5101)
Level 05 Hops: Accuracy 11.89% (22/185)
Level 06 Hops: Accuracy 04.76% (5/105)
Level 07 Hops: Accuracy 03.23% (5/155)
Level 08 Hops: Accuracy 09.63% (13/135)

Level 09 Hops: Accuracy 06.45% (8/124)
Level 10 Hops: Accuracy 03.28% (4/122)

Overall Dataset Accuracy: 13.77%
Overall Average Fitness: 0.1231

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16131 (100.0%)
Exactly 2-Word Answers: 0 (0.0%)
==========
==========
🎯 STRATIFIED BASELINE REPORT t=2.0
==========
Execution Time: 146.09 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 13.49% (688/5101)
Level 03 Hops: Accuracy 15.50% (791/5103)
Level 04 Hops: Accuracy 13.90% (709/5101)
Level 05 Hops: Accuracy 10.27% (19/185)
Level 06 Hops: Accuracy 05.71% (6/105)
Level 07 Hops: Accuracy 03.87% (6/155)
Level 08 Hops: Accuracy 08.89% (12/135)
Level 09 Hops: Accuracy 07.26% (9/124)
Level 10 Hops: Accuracy 01.64% (2/122)

Overall Dataset Accuracy: 13.90%
Overall Average Fitness: 0.1241

Average Answer Length: 1.02 words
Exactly 1-Word Answers: 16096 (99.8%)
Exactly 2-Word Answers: 4 (0.0%)
==========

==========
🎯 STRATIFIED FEW SHOTS REPORT
==========
Execution Time: 113.64 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 24.49% (1249/5101)
Level 03 Hops: Accuracy 21.48% (1096/5103)
Level 04 Hops: Accuracy 18.17% (927/5101)
Level 05 Hops: Accuracy 15.14% (28/185)
Level 06 Hops: Accuracy 13.33% (14/105)
Level 07 Hops: Accuracy 09.03% (14/155)
Level 08 Hops: Accuracy 12.59% (17/135)
Level 09 Hops: Accuracy 10.48% (13/124)

Level 10 Hops: Accuracy 04.92% (6/122)

Overall Dataset Accuracy: 20.85%
Overall Average Fitness: 0.1773

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16115 (99.9%)
Exactly 2-Word Answers: 11 (0.1%)
==========
==========
🎯 STRATIFIED FEW SHOTS REPORT
==========
Execution Time: 120.00 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 25.39% (1295/5101)
Level 03 Hops: Accuracy 21.75% (1110/5103)
Level 04 Hops: Accuracy 18.64% (951/5101)
Level 05 Hops: Accuracy 14.59% (27/185)
Level 06 Hops: Accuracy 15.24% (16/105)
Level 07 Hops: Accuracy 09.03% (14/155)
Level 08 Hops: Accuracy 11.85% (16/135)
Level 09 Hops: Accuracy 11.29% (14/124)
Level 10 Hops: Accuracy 05.74% (7/122)

Overall Dataset Accuracy: 21.39%
Overall Average Fitness: 0.1711

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16129 (100.0%)
Exactly 2-Word Answers: 0 (0.0%)
==========
==========
🎯 STRATIFIED FEW SHOTS REPORT
==========
Execution Time: 135.50 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 26.29% (1341/5101)
Level 03 Hops: Accuracy 22.38% (1142/5103)
Level 04 Hops: Accuracy 18.96% (967/5101)
Level 05 Hops: Accuracy 14.05% (26/185)
Level 06 Hops: Accuracy 14.29% (15/105)
Level 07 Hops: Accuracy 09.68% (15/155)
Level 08 Hops: Accuracy 12.59% (17/135)
Level 09 Hops: Accuracy 08.87% (11/124)
Level 10 Hops: Accuracy 05.74% (7/122)

Overall Dataset Accuracy: 21.95%
Overall Average Fitness: 0.1756

Average Answer Length: 1.00 words
Exactly 1-Word Answers: 16130 (100.0%)
Exactly 2-Word Answers: 1 (0.0%)
==========

==========

🎯 STRATIFIED FEW SHOTS REPORT

Level 02 Hops: Accuracy 23.92% (1220/5101)
Level 03 Hops: Accuracy 20.18% (1030/5103)
Level 04 Hops: Accuracy 19.15% (977/5101)
Level 05 Hops: Accuracy 16.76% (31/185)
Level 06 Hops: Accuracy 26.67% (28/105)
Level 07 Hops: Accuracy 17.42% (27/155)
Level 08 Hops: Accuracy 16.30% (22/135)
Level 09 Hops: Accuracy 21.77% (27/124)
Level 10 Hops: Accuracy 20.49% (25/122)

Overall Dataset Accuracy: 21.00%
Overall Average Fitness: 0.1680
Execution Time: 149.59 seconds

==========
🎯 STRATIFIED BEST ERA INDIVIDUAL REPORT (with species protection)
==========
Execution Time: 1932.95 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 23.76% (1212/5101)
Level 03 Hops: Accuracy 18.13% (925/5103)
Level 04 Hops: Accuracy 18.82% (960/5101)
Level 05 Hops: Accuracy 17.84% (33/185)
Level 06 Hops: Accuracy 14.29% (15/105)
Level 07 Hops: Accuracy 11.61% (18/155)
Level 08 Hops: Accuracy 14.81% (20/135)
Level 09 Hops: Accuracy 13.71% (17/124)
Level 10 Hops: Accuracy 10.66% (13/122)

Overall Dataset Accuracy: 19.92%
Overall Average Fitness: 0.1693

Average Answer Length: 4.80 words
Exactly 1-Word Answers: 15 (0.1%)
Exactly 2-Word Answers: 12 (0.1%)

==========
==========
🎯 STRATIFIED BEST ERA INDIVIDUAL REPORT (with species protection)
+ added node to enforce 1 word answers:

```
"Review the previous reasoning. Extract the exact final family
 relationship. Output ONLY that single word, with no punctuation or
 explanation."
```

==========
Execution Time: 2075.11 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 13.00% (663/5101)
Level 03 Hops: Accuracy 14.70% (750/5103)
Level 04 Hops: Accuracy 15.70% (801/5101)
Level 05 Hops: Accuracy 09.73% (18/185)
Level 06 Hops: Accuracy 04.76% (5/105)
Level 07 Hops: Accuracy 04.52% (7/155)
Level 08 Hops: Accuracy 06.67% (9/135)
Level 09 Hops: Accuracy 08.87% (11/124)
Level 10 Hops: Accuracy 06.56% (8/122)

Overall Dataset Accuracy: 14.08%
Overall Average Fitness: 0.1197

Average Answer Length: 1.53 words
Exactly 1-Word Answers: 15933 (98.8%)
Exactly 2-Word Answers: 26 (0.2%)
==========

==========
🎯 STRATIFIED BEST ERA INDIVIDUAL REPORT (no species protection) balanced subset
==========
Execution Time: 1202.62 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 17.55% (895/5101)
Level 03 Hops: Accuracy 17.24% (880/5103)
Level 04 Hops: Accuracy 15.51% (791/5101)
Level 05 Hops: Accuracy 11.89% (22/185)
Level 06 Hops: Accuracy 10.48% (11/105)
Level 07 Hops: Accuracy 07.10% (11/155)
Level 08 Hops: Accuracy 08.15% (11/135)
Level 09 Hops: Accuracy 09.68% (12/124)
Level 10 Hops: Accuracy 10.66% (13/122)

Overall Dataset Accuracy: 16.40%

Overall Average Fitness: 0.1312

Average Answer Length: 1.02 words
Exactly 1-Word Answers: 15915 (98.7%)
Exactly 2-Word Answers: 153 (0.9%)
==========

==========
🎯 STRATIFIED BASELINE REPORT
(6 hops subset) (best accuracy individual)
==========
Execution Time: 1634.23 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 17.11% (873/5101)
Level 03 Hops: Accuracy 18.83% (961/5103)
Level 04 Hops: Accuracy 17.70% (903/5101)
Level 05 Hops: Accuracy 14.59% (27/185)
Level 06 Hops: Accuracy 16.19% (17/105)
Level 07 Hops: Accuracy 07.74% (12/155)
Level 08 Hops: Accuracy 12.59% (17/135)
Level 09 Hops: Accuracy 08.87% (11/124)
Level 10 Hops: Accuracy 07.38% (9/122)

Overall Dataset Accuracy: 17.54%
Overall Average Fitness: 0.1490

Average Answer Length: 1.03 words
Exactly 1-Word Answers: 15790 (97.9%)
Exactly 2-Word Answers: 191 (1.2%)
==========
==========
🎯 STRATIFIED BASELINE REPORT
(6 hops subset) (best fitness individual)
==========
Execution Time: 1454.01 seconds
Total Problems Evaluated: 16131

Level 02 Hops: Accuracy 16.84% (859/5101)
Level 03 Hops: Accuracy 18.03% (920/5103)
Level 04 Hops: Accuracy 17.78% (907/5101)
Level 05 Hops: Accuracy 16.76% (31/185)
Level 06 Hops: Accuracy 13.33% (14/105)
Level 07 Hops: Accuracy 10.32% (16/155)
Level 08 Hops: Accuracy 17.04% (23/135)
Level 09 Hops: Accuracy 12.10% (15/124)
Level 10 Hops: Accuracy 06.56% (8/122)

Overall Dataset Accuracy: 17.31%
Overall Average Fitness: 0.1472

Average Answer Length: 1.03 words
Exactly 1-Word Answers: 15802 (98.0%)
Exactly 2-Word Answers: 148 (0.9%)
==========

Graph length tests:
1D TESTS:
🚀 Starting Cognitive Event Horizon Test | MODE: 1D
Testing depths: [2, 4, 5, 7, 10, 15] with 20 trials each.

--- Running Tests for Depth: 2 ---
  Trial 01/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 02/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 03/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 04/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 05/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 06/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 07/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 08/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 09/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 10/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 11/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 12/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 13/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 14/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 15/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 16/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 17/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 18/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 19/20 | Expected: 0 | Got: Last Position: 0 | ✅ PASS
  Trial 20/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
➡️ Depth 2 Completed. Average Success: 100.0%

--- Running Tests for Depth: 4 ---
  Trial 01/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 02/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
  Trial 03/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 04/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 05/20 | Expected: -4 | Got: Last Position: -4 | ✅ PASS
  Trial 06/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
  Trial 07/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 08/20 | Expected: -2 | Got: Last Position: -2 | ✅ PASS
  Trial 09/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
  Trial 10/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS

Trial 11/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 12/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 13/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 14/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 15/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
Trial 16/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
Trial 17/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 18/20 | Expected: 2 | Got: Last Position: 2 | ✅ PASS
Trial 19/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
Trial 20/20 | Expected: 0 | Got: Starting Position: 0 | ✅ PASS
➡️ Depth 4 Completed. Average Success: 100.0%

--- Running Tests for Depth: 5 ---
Trial 01/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 02/20 | Expected: -1 | Got: Last Position: -1 | ✅ PASS
Trial 03/20 | Expected: -1 | Got: Last Position: -1 | ✅ PASS
Trial 04/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 05/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 06/20 | Expected: -1 | Got: Last Position: -3 | ❌ FAIL
Trial 07/20 | Expected: 1 | Got: Last Position: -1 | ❌ FAIL
Trial 08/20 | Expected: -5 | Got: Last Position: -5 | ✅ PASS
Trial 09/20 | Expected: -1 | Got: Last Position: -1 | ✅ PASS
Trial 10/20 | Expected: -3 | Got: Last Position: -3 | ✅ PASS
Trial 11/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 12/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 13/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 14/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 15/20 | Expected: -1 | Got: Last Position: -1 | ✅ PASS
Trial 16/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 17/20 | Expected: -3 | Got: Last Position: -3 | ✅ PASS
Trial 18/20 | Expected: -3 | Got: Last Position: -3 | ✅ PASS
Trial 19/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 20/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
➡️ Depth 5 Completed. Average Success: 90.0%

--- Running Tests for Depth: 7 ---
Trial 01/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 02/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 03/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 04/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 05/20 | Expected: -3 | Got: -3 | ✅ PASS
Trial 06/20 | Expected: -3 | Got: Last Position: -5 | ❌ FAIL
Trial 07/20 | Expected: -7 | Got: Last Position: -7 | ✅ PASS
Trial 08/20 | Expected: 1 | Got: -3 | ❌ FAIL
Trial 09/20 | Expected: 5 | Got: 5 | ✅ PASS
Trial 10/20 | Expected: -5 | Got: Last Position: -5 | ✅ PASS
Trial 11/20 | Expected: 1 | Got: 1 | ✅ PASS
Trial 12/20 | Expected: -3 | Got: -3 | ✅ PASS

Trial 13/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 14/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 15/20 | Expected: 1 | Got: 1 | ✅ PASS
Trial 16/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
Trial 17/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 18/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 19/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 20/20 | Expected: 1 | Got: Last Position: 1 | ✅ PASS
➡️ Depth 7 Completed. Average Success: 90.0%

--- Running Tests for Depth: 10 ---
Trial 01/20 | Expected: 0 | Got: -2 | ❌ FAIL
Trial 02/20 | Expected: 4 | Got: 4 | ✅ PASS
Trial 03/20 | Expected: -2 | Got: -2 | ✅ PASS
Trial 04/20 | Expected: -2 | Got: -2 | ✅ PASS
Trial 05/20 | Expected: 4 | Got: 4 | ✅ PASS
Trial 06/20 | Expected: 0 | Got: -2 | ❌ FAIL
Trial 07/20 | Expected: 2 | Got: 2 | ✅ PASS
Trial 08/20 | Expected: 2 | Got: 2 | ✅ PASS
Trial 09/20 | Expected: 6 | Got: 6 | ✅ PASS
Trial 10/20 | Expected: 2 | Got: 2 | ✅ PASS
Trial 11/20 | Expected: 0 | Got: 0 | ✅ PASS
Trial 12/20 | Expected: -2 | Got: -4 | ❌ FAIL
Trial 13/20 | Expected: 0 | Got: -2 | ❌ FAIL
Trial 14/20 | Expected: -2 | Got: -4 | ❌ FAIL
Trial 15/20 | Expected: 2 | Got: 2 | ✅ PASS
Trial 16/20 | Expected: -2 | Got: -4 | ❌ FAIL
Trial 17/20 | Expected: 0 | Got: -2 | ❌ FAIL
Trial 18/20 | Expected: 0 | Got: 0 | ✅ PASS
Trial 19/20 | Expected: -4 | Got: -4 | ✅ PASS
Trial 20/20 | Expected: 0 | Got: 0 | ✅ PASS
➡️ Depth 10 Completed. Average Success: 65.0%

--- Running Tests for Depth: 15 ---
Trial 01/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 02/20 | Expected: -3 | Got: -3 | ✅ PASS
Trial 03/20 | Expected: 5 | Got: 5 | ✅ PASS
Trial 04/20 | Expected: 7 | Got: 7 | ✅ PASS
Trial 05/20 | Expected: -1 | Got: 1 | ❌ FAIL
Trial 06/20 | Expected: 5 | Got: -3 | ❌ FAIL
Trial 07/20 | Expected: 1 | Got: 1 | ✅ PASS
Trial 08/20 | Expected: -3 | Got: -5 | ❌ FAIL
Trial 09/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 10/20 | Expected: 1 | Got: 1 | ✅ PASS
Trial 11/20 | Expected: -5 | Got: -5 | ✅ PASS
Trial 12/20 | Expected: 3 | Got: -3 | ❌ FAIL
Trial 13/20 | Expected: 1 | Got: -3 | ❌ FAIL
Trial 14/20 | Expected: -3 | Got: -3 | ✅ PASS

Trial 15/20 | Expected: -1 | Got: -1 | ✅ PASS
Trial 16/20 | Expected: -3 | Got: -3 | ✅ PASS
Trial 17/20 | Expected: -7 | Got: -7 | ✅ PASS
Trial 18/20 | Expected: 1 | Got: -1 | ❌ FAIL
Trial 19/20 | Expected: 3 | Got: 3 | ✅ PASS
Trial 20/20 | Expected: 5 | Got: 5 | ✅ PASS
➡️ Depth 15 Completed. Average Success: 70.0%

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🏆 FINAL COGNITIVE EVENT HORIZON REPORT | 1D MODE
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Graph Depth	Avg Context Tokens	Success Rate
2	46	100.0%
4	70	100.0%
5	82	90.0%
7	106	90.0%
10	142	65.0%
15	203	70.0%

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2D TESTS:
🚀 Starting Cognitive Event Horizon Test | MODE: 2D
Testing depths: [2, 4, 5, 7, 10, 15] with 20 trials each.

--- Running Tests for Depth: 2 ---
Trial 01/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 02/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 03/20 | Expected: (1, -1) | Got: Last Position: (1, -1) | ✅ PASS
Trial 04/20 | Expected: (-1, 1) | Got: Last Position: (-1, 1) | ✅ PASS
Trial 05/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 06/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 07/20 | Expected: (-1, -1) | Got: Last Position: (-1, -1) | ✅ PASS
Trial 08/20 | Expected: (2, 0) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 09/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 10/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 11/20 | Expected: (0, 2) | Got: Last Position: (0, 2) | ✅ PASS
Trial 12/20 | Expected: (1, -1) | Got: Last Position: (1, -1) | ✅ PASS
Trial 13/20 | Expected: (-1, -1) | Got: Last Position: (-1, -1) | ✅ PASS
Trial 14/20 | Expected: (0, 2) | Got: Last Position: (0, 2) | ✅ PASS
Trial 15/20 | Expected: (-2, 0) | Got: Last Position: (-1, 0) | ❌ FAIL
Trial 16/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 17/20 | Expected: (-2, 0) | Got: Last Position: (-1, 0) | ❌ FAIL
Trial 18/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 19/20 | Expected: (1, 1) | Got: Last Position: (1, 1) | ✅ PASS

Trial 20/20 | Expected: (1, -1) | Got: Last Position: (1, -1) | ✅ PASS
➡️ Depth 2 Completed. Average Success: 85.0%

--- Running Tests for Depth: 4 ---
Trial 01/20 | Expected: (0, 0) | Got: Last Position: (0, 0) | ✅ PASS
Trial 02/20 | Expected: (-2, 0) | Got: (-1, -1) | ❌ FAIL
Trial 03/20 | Expected: (0, 0) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 04/20 | Expected: (-4, 0) | Got: Last Position: (-2, 0) | ❌ FAIL
Trial 05/20 | Expected: (-1, 1) | Got: Last Position: (-1, 1) | ✅ PASS
Trial 06/20 | Expected: (-2, 0) | Got: Last Position: (-2, 0) | ✅ PASS
Trial 07/20 | Expected: (4, 0) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 08/20 | Expected: (1, 3) | Got: Starting Position: (1, 3) | ✅ PASS
Trial 09/20 | Expected: (1, 1) | Got: Last Position: (0, 2) | ❌ FAIL
Trial 10/20 | Expected: (0, 0) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 11/20 | Expected: (0, 0) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 12/20 | Expected: (1, 1) | Got: Last Position: (2, 1) | ❌ FAIL
Trial 13/20 | Expected: (-1, 1) | Got: Last Position: (-2, 2) | ❌ FAIL
Trial 14/20 | Expected: (2, 0) | Got: Last Position: (2, 0) | ✅ PASS
Trial 15/20 | Expected: (0, 0) | Got: Starting Position: (0, 0) | ✅ PASS
Trial 16/20 | Expected: (1, -1) | Got: Last Position: (1, -1) | ✅ PASS
Trial 17/20 | Expected: (-3, -1) | Got: Last Position: (-3, -1) | ✅ PASS
Trial 18/20 | Expected: (1, -1) | Got: Last Position: (1, -1) | ✅ PASS
Trial 19/20 | Expected: (-1, 1) | Got: Starting Position: (0, 0) | ❌ FAIL
Trial 20/20 | Expected: (2, 0) | Got: Last Position: (1, 1) | ❌ FAIL
➡️ Depth 4 Completed. Average Success: 45.0%

--- Running Tests for Depth: 5 ---
Trial 01/20 | Expected: (1, -2) | Got: Last Position: (1, -2) | ✅ PASS
Trial 02/20 | Expected: (-1, -2) | Got: (-1, -1) | ❌ FAIL
Trial 03/20 | Expected: (-3, 0) | Got: Last Position: (0, -1) | ❌ FAIL
Trial 04/20 | Expected: (0, -3) | Got: Last Position: (0, -3) | ✅ PASS
Trial 05/20 | Expected: (3, 2) | Got: Starting Position: (2, 3) | ❌ FAIL
Trial 06/20 | Expected: (-1, 0) | Got: Last Position: (-2, 1) | ❌ FAIL
Trial 07/20 | Expected: (1, 0) | Got: (1, 0) | ✅ PASS
Trial 08/20 | Expected: (-1, 0) | Got: Last Position: (-1, -1) | ❌ FAIL
Trial 09/20 | Expected: (2, -1) | Got: Last Position: (2, -1) | ✅ PASS
Trial 10/20 | Expected: (0, -1) | Got: Last Position: (1, -1) | ❌ FAIL
Trial 11/20 | Expected: (-1, 2) | Got: (-1, 2) | ✅ PASS
Trial 12/20 | Expected: (3, 0) | Got: Last Position: (2, 1) | ❌ FAIL
Trial 13/20 | Expected: (0, 1) | Got: Last Position: (0, 1) | ✅ PASS
Trial 14/20 | Expected: (1, 0) | Got: Last Position: (2, -1) | ❌ FAIL
Trial 15/20 | Expected: (0, 1) | Got: Last Position: (1, 0) | ❌ FAIL
Trial 16/20 | Expected: (2, 1) | Got: Last Position: (2, 0) | ❌ FAIL
Trial 17/20 | Expected: (4, -1) | Got: Last Position: (2, -1) | ❌ FAIL
Trial 18/20 | Expected: (1, -2) | Got: Last Position: (0, -1) | ❌ FAIL
Trial 19/20 | Expected: (2, -3) | Got: Last Position: (2, -3) | ✅ PASS
Trial 20/20 | Expected: (3, 0) | Got: (2, 1) | ❌ FAIL
➡️ Depth 5 Completed. Average Success: 35.0%

--- Running Tests for Depth: 7 ---
  Trial 01/20 | Expected: (0, -1) | Got: Last Position: (-1, -1) | ❌ FAIL
  Trial 02/20 | Expected: (4, 1) | Got: Last Position: (5, 1) | ❌ FAIL
  Trial 03/20 | Expected: (0, 1) | Got: (0, 1) | ✅ PASS
  Trial 04/20 | Expected: (-2, -3) | Got: Last Position: (-2, -3) | ✅ PASS
  Trial 05/20 | Expected: (0, 1) | Got: Last Position: (-2, 2) | ❌ FAIL
  Trial 06/20 | Expected: (-1, 2) | Got: Last Position: (1, 1) | ❌ FAIL
  Trial 07/20 | Expected: (0, 1) | Got: (1, 0) | ❌ FAIL
  Trial 08/20 | Expected: (-2, -3) | Got: Last Position: (-2, -2) | ❌ FAIL
  Trial 09/20 | Expected: (1, -2) | Got: Last Position: (2, -2) | ❌ FAIL
  Trial 10/20 | Expected: (3, -2) | Got: Last Position: (2, -1) | ❌ FAIL
  Trial 11/20 | Expected: (2, -1) | Got: Last Position: (2, -1) | ✅ PASS
  Trial 12/20 | Expected: (0, -1) | Got: Last Position: (1, -1) | ❌ FAIL
  Trial 13/20 | Expected: (2, 1) | Got: (2, 1) | ✅ PASS
  Trial 14/20 | Expected: (-2, -1) | Got: Last Position: (-2, -2) | ❌ FAIL
  Trial 15/20 | Expected: (-2, -1) | Got: Last Position: (1, -1) | ❌ FAIL
  Trial 16/20 | Expected: (5, 2) | Got: Last Position: (2, 1) | ❌ FAIL
  Trial 17/20 | Expected: (3, 0) | Got: (3, 0) | ✅ PASS
  Trial 18/20 | Expected: (-2, -1) | Got: (-1, -2) | ❌ FAIL
  Trial 19/20 | Expected: (0, -3) | Got: Last Position: (1, 0) | ❌ FAIL
  Trial 20/20 | Expected: (2, -1) | Got: Last Position: (3, -2) | ❌ FAIL
➡️ Depth 7 Completed. Average Success: 25.0%

--- Running Tests for Depth: 10 ---
  Trial 01/20 | Expected: (-2, 0) | Got: (0, 1) | ❌ FAIL
  Trial 02/20 | Expected: (-2, 4) | Got: (1, 1) | ❌ FAIL
  Trial 03/20 | Expected: (2, 0) | Got: (2, 0) | ✅ PASS
  Trial 04/20 | Expected: (2, 4) | Got: (1, 5) | ❌ FAIL
  Trial 05/20 | Expected: (-2, 0) | Got: (0, -1) | ❌ FAIL
  Trial 06/20 | Expected: (1, 3) | Got: (1, 0) | ❌ FAIL
  Trial 07/20 | Expected: (2, 0) | Got: (2, 1) | ❌ FAIL
  Trial 08/20 | Expected: (-2, 2) | Got: (-2, 1) | ❌ FAIL
  Trial 09/20 | Expected: (3, -1) | Got: (3, -1) | ✅ PASS
  Trial 10/20 | Expected: (2, 0) | Got: (-1, 3) | ❌ FAIL
  Trial 11/20 | Expected: (3, -3) | Got: (3, -3) | ✅ PASS
  Trial 12/20 | Expected: (2, 0) | Got: (-1, 3) | ❌ FAIL
  Trial 13/20 | Expected: (-3, 1) | Got: (-2, 0) | ❌ FAIL
  Trial 14/20 | Expected: (-3, 3) | Got: (0, 0) | ❌ FAIL
  Trial 15/20 | Expected: (-3, 1) | Got: (5, 3) | ❌ FAIL
  Trial 16/20 | Expected: (1, -1) | Got: (1, -2) | ❌ FAIL
  Trial 17/20 | Expected: (-1, 3) | Got: Last Position: (0, 0) | ❌ FAIL
  Trial 18/20 | Expected: (-2, 4) | Got: (0, 2) | ❌ FAIL
  Trial 19/20 | Expected: (0, 2) | Got: (3, 0) | ❌ FAIL
  Trial 20/20 | Expected: (-1, -3) | Got: (0, -3) | ❌ FAIL
➡️ Depth 10 Completed. Average Success: 15.0%

--- Running Tests for Depth: 15 ---

Trial 01/20 | Expected: (0, 1) | Got: (1, 1) | ❌ FAIL
Trial 02/20 | Expected: (0, -3) | Got: (-2, -3) | ❌ FAIL
Trial 03/20 | Expected: (1, 0) | Got: (1, 1) | ❌ FAIL
Trial 04/20 | Expected: (0, 1) | Got: (1, 0) | ❌ FAIL
Trial 05/20 | Expected: (5, 0) | Got: (3, 0) | ❌ FAIL
Trial 06/20 | Expected: (-1, 2) | Got: (1, 1) | ❌ FAIL
Trial 07/20 | Expected: (2, -5) | Got: (1, -4) | ❌ FAIL
Trial 08/20 | Expected: (-4, 5) | Got: (0, -2) | ❌ FAIL
Trial 09/20 | Expected: (-4, 3) | Got: (0, -1) | ❌ FAIL
Trial 10/20 | Expected: (-4, 1) | Got: (0, -5) | ❌ FAIL
Trial 11/20 | Expected: (-6, 1) | Got: (-4, -4) | ❌ FAIL
Trial 12/20 | Expected: (0, -1) | Got: (1, -3) | ❌ FAIL
Trial 13/20 | Expected: (2, 3) | Got: (2, 1) | ❌ FAIL
Trial 14/20 | Expected: (2, 1) | Got: (2, 1) | ✅ PASS
Trial 15/20 | Expected: (0, -3) | Got: (-1, -2) | ❌ FAIL
Trial 16/20 | Expected: (4, 3) | Got: (1, 1) | ❌ FAIL
Trial 17/20 | Expected: (-1, 0) | Got: (-2, 1) | ❌ FAIL
Trial 18/20 | Expected: (2, -3) | Got: (1, -2) | ❌ FAIL
Trial 19/20 | Expected: (-2, 5) | Got: (3, 6) | ❌ FAIL
Trial 20/20 | Expected: (2, 7) | Got: (1, 7) | ❌ FAIL
➡️ Depth 15 Completed. Average Success: 5.0%

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🏆 FINAL COGNITIVE EVENT HORIZON REPORT | 2D MODE
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Graph Depth	Avg Context Tokens	Success Rate
2	48	85.0%
4	68	45.0%
5	77	35.0%
7	97	25.0%
10	129	15.0%
15	173	5.0%

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3D TESTS
🚀 Starting Cognitive Event Horizon Test | MODE: 3D
Testing depths: [2, 4, 5, 7, 10, 15] with 20 trials each.

--- Running Tests for Depth: 2 ---
Trial 01/20 | Expected: (0, 0, 0) | Got: Starting Position: (0, 0, | ✅ PASS
Trial 02/20 | Expected: (0, 0, 0) | Got: Starting Position: (0, 0, | ✅ PASS
Trial 03/20 | Expected: (0, -1, -1) | Got: Last Position: (0, -1, -1 | ✅ PASS
Trial 04/20 | Expected: (-2, 0, 0) | Got: Last Position: (-2, 0, 0) | ✅ PASS
Trial 05/20 | Expected: (0, -2, 0) | Got: Starting Position: (0, 0, | ❌ FAIL

Trial 06/20 | Expected: (-1, 0, 1) | Got: Last Position: (-1, 0, 1) | ✅ PASS
Trial 07/20 | Expected: (1, -1, 0) | Got: Last Position: (1, 1, -1) | ❌ FAIL
Trial 08/20 | Expected: (-1, 0, -1) | Got: Last Position: (-1, 0, -1 | ✅ PASS
Trial 09/20 | Expected: (0, 0, -2) | Got: Starting Position: (0, 0, | ❌ FAIL
Trial 10/20 | Expected: (0, -1, -1) | Got: Last Position: (0, -1, -1 | ✅ PASS
Trial 11/20 | Expected: (-1, 1, 0) | Got: Last Position: (-1, 0, 1) | ❌ FAIL
Trial 12/20 | Expected: (-1, -1, 0) | Got: Last Position: (-1, -1, 0 | ✅ PASS
Trial 13/20 | Expected: (1, 0, -1) | Got: Last Position: (1, 1, -1) | ❌ FAIL
Trial 14/20 | Expected: (-2, 0, 0) | Got: Last Position: (-2, 0, 0) | ✅ PASS
Trial 15/20 | Expected: (0, 0, 0) | Got: Starting Position: (0, 0, | ✅ PASS
Trial 16/20 | Expected: (0, -1, -1) | Got: Last Position: (0, -1, -1 | ✅ PASS
Trial 17/20 | Expected: (0, -1, -1) | Got: Last Position: (0, 0, -1) | ❌ FAIL
Trial 18/20 | Expected: (0, 1, -1) | Got: Starting Position: (0, 0, | ❌ FAIL
Trial 19/20 | Expected: (1, 1, 0) | Got: Last Position: (1, 0, 1) | ❌ FAIL
Trial 20/20 | Expected: (1, 0, 1) | Got: Last Position: (1, 0, 1) | ✅ PASS
➡️ Depth 2 Completed. Average Success: 60.0%

--- Running Tests for Depth: 4 ---
Trial 01/20 | Expected: (1, -2, -1) | Got: (1, -1, -1) | ❌ FAIL
Trial 02/20 | Expected: (0, 0, 0) | Got: Starting Position: (0, 0, | ✅ PASS
Trial 03/20 | Expected: (1, 0, 1) | Got: (0, 1, 1) | ❌ FAIL
Trial 04/20 | Expected: (1, 0, 1) | Got: (1, 0, 1) | ✅ PASS
Trial 05/20 | Expected: (0, 0, 2) | Got: Starting Position: (0, 0, | ❌ FAIL
Trial 06/20 | Expected: (0, 1, -1) | Got: (0, 1, -1) | ✅ PASS
Trial 07/20 | Expected: (-1, -1, 0) | Got: (-1, -1, 0) | ✅ PASS
Trial 08/20 | Expected: (0, 0, 0) | Got: (0, 0, 0) | ✅ PASS
Trial 09/20 | Expected: (-2, 0, 0) | Got: (-2, 0, 0) | ✅ PASS
Trial 10/20 | Expected: (1, 0, -1) | Got: (1, 0, 0) | ❌ FAIL
Trial 11/20 | Expected: (1, -1, 0) | Got: (1, -2, 1) | ❌ FAIL
Trial 12/20 | Expected: (0, 1, 1) | Got: Starting Position: (0, 0, | ❌ FAIL
Trial 13/20 | Expected: (-1, -2, 1) | Got: (-1, -2, 1) | ✅ PASS
Trial 14/20 | Expected: (-1, 1, -2) | Got: (-1, 0, -2) | ❌ FAIL
Trial 15/20 | Expected: (1, 1, -2) | Got: (1, 0, -1) | ❌ FAIL
Trial 16/20 | Expected: (1, -2, 1) | Got: (1, -1, 0) | ❌ FAIL
Trial 17/20 | Expected: (4, 0, 0) | Got: (1, 1, 0) | ❌ FAIL
Trial 18/20 | Expected: (1, 1, -2) | Got: Starting Position: (1, 0, | ❌ FAIL
Trial 19/20 | Expected: (0, 0, 0) | Got: (-1, 0, 0) | ❌ FAIL
Trial 20/20 | Expected: (1, -1, 0) | Got: (1, -1, 0) | ✅ PASS
➡️ Depth 4 Completed. Average Success: 40.0%

--- Running Tests for Depth: 5 ---
Trial 01/20 | Expected: (1, 2, 0) | Got: (1, 1, 1) | ❌ FAIL
Trial 02/20 | Expected: (3, 0, 0) | Got: (2, 0, 1) | ❌ FAIL
Trial 03/20 | Expected: (0, 0, -1) | Got: (0, 0, -1) | ✅ PASS
Trial 04/20 | Expected: (-1, -1, 1) | Got: (-2, -1, 1) | ❌ FAIL
Trial 05/20 | Expected: (1, -2, 0) | Got: (1, -2, 0) | ✅ PASS
Trial 06/20 | Expected: (0, 0, 1) | Got: (0, 0, 1) | ✅ PASS
Trial 07/20 | Expected: (-1, 0, 0) | Got: (-2, 0, 0) | ❌ FAIL

Trial 08/20 | Expected: (2, 0, 1) | Got: (2, -1, 2) | ❌ FAIL
Trial 09/20 | Expected: (-2, 0, 1) | Got: (-2, -1, 2) | ❌ FAIL
Trial 10/20 | Expected: (-2, 1, -2) | Got: (-2, 0, -1) | ❌ FAIL
Trial 11/20 | Expected: (-2, 2, -1) | Got: (-2, 0, 1) | ❌ FAIL
Trial 12/20 | Expected: (0, 0, -1) | Got: (0, 0, 0) | ❌ FAIL
Trial 13/20 | Expected: (1, 0, 0) | Got: (1, 0, 0) | ✅ PASS
Trial 14/20 | Expected: (0, 0, 1) | Got: (0, 0, 1) | ✅ PASS
Trial 15/20 | Expected: (0, 0, 1) | Got: (0, 0, 1) | ✅ PASS
Trial 16/20 | Expected: (0, 0, -1) | Got: (0, 0, 0) | ❌ FAIL
Trial 17/20 | Expected: (0, -1, 0) | Got: (0, 0, 0) | ❌ FAIL
Trial 18/20 | Expected: (0, 0, 1) | Got: (0, 0, 1) | ✅ PASS
Trial 19/20 | Expected: (-1, 2, 0) | Got: (-1, 1, 1) | ❌ FAIL
Trial 20/20 | Expected: (0, 0, 1) | Got: (0, 0, 1) | ✅ PASS
➡️ Depth 5 Completed. Average Success: 40.0%

--- Running Tests for Depth: 7 ---
Trial 01/20 | Expected: (0, 1, 2) | Got: (0, -1, 4) | ❌ FAIL
Trial 02/20 | Expected: (-1, -1, -3) | Got: (-1, -4, 0) | ❌ FAIL
Trial 03/20 | Expected: (-2, 0, 1) | Got: (-2, -1, 2) | ❌ FAIL
Trial 04/20 | Expected: (-1, 1, 1) | Got: (0, 0, 1) | ❌ FAIL
Trial 05/20 | Expected: (-1, -2, 2) | Got: (0, -1, 0) | ❌ FAIL
Trial 06/20 | Expected: (3, 0, 0) | Got: (2, 1, 0) | ❌ FAIL
Trial 07/20 | Expected: (0, -1, -2) | Got: (0, 0, -2) | ❌ FAIL
Trial 08/20 | Expected: (0, -2, -1) | Got: (0, 0, -2) | ❌ FAIL
Trial 09/20 | Expected: (-1, 0, 0) | Got: (-1, 0, 0) | ✅ PASS
Trial 10/20 | Expected: (0, 1, 0) | Got: (-2, 0, 0) | ❌ FAIL
Trial 11/20 | Expected: (3, -2, 0) | Got: (2, -2, 1) | ❌ FAIL
Trial 12/20 | Expected: (2, 0, 1) | Got: (1, 0, 0) | ❌ FAIL
Trial 13/20 | Expected: (0, -1, 0) | Got: (0, -1, 0) | ✅ PASS
Trial 14/20 | Expected: (2, 1, 2) | Got: (-1, 0, 3) | ❌ FAIL
Trial 15/20 | Expected: (-1, 3, 1) | Got: (-1, 0, 3) | ❌ FAIL
Trial 16/20 | Expected: (1, -2, -2) | Got: (2, -2, -2) | ❌ FAIL
Trial 17/20 | Expected: (0, 3, 0) | Got: (0, 0, -1) | ❌ FAIL
Trial 18/20 | Expected: (-3, 2, 0) | Got: (-4, 1, 1) | ❌ FAIL
Trial 19/20 | Expected: (-1, 1, 1) | Got: (-1, 0, 2) | ❌ FAIL
Trial 20/20 | Expected: (2, -1, -2) | Got: (1, 0, -3) | ❌ FAIL
➡️ Depth 7 Completed. Average Success: 10.0%

--- Running Tests for Depth: 10 ---
Trial 01/20 | Expected: (-1, 0, 1) | Got: (-2, -1, 2) | ❌ FAIL
Trial 02/20 | Expected: (-1, 4, -1) | Got: (-1, 1, 2) | ❌ FAIL
Trial 03/20 | Expected: (-2, 1, 1) | Got: (-2, 1, 1) | ✅ PASS
Trial 04/20 | Expected: (-1, 3, 0) | Got: (-4, 0, 3) | ❌ FAIL
Trial 05/20 | Expected: (-1, 0, -1) | Got: (1, -1, 0) | ❌ FAIL
Trial 06/20 | Expected: (-1, -3, -2) | Got: (-2, -3, -1) | ❌ FAIL
Trial 07/20 | Expected: (-1, 1, 0) | Got: (-1, -1, 1) | ❌ FAIL
Trial 08/20 | Expected: (-2, 0, 4) | Got: (-2, -1, 5) | ❌ FAIL
Trial 09/20 | Expected: (-3, -3, 0) | Got: (-4, -4, 0) | ❌ FAIL

Trial 10/20 | Expected: (-1, -2, -3) | Got: (0, 0, -1) | ❌ FAIL
Trial 11/20 | Expected: (-2, -1, -1) | Got: (-2, -3, 1) | ❌ FAIL
Trial 12/20 | Expected: (-1, 0, 1) | Got: (-4, 1, 1) | ❌ FAIL
Trial 13/20 | Expected: (2, 1, -1) | Got: (1, 1, -1) | ❌ FAIL
Trial 14/20 | Expected: (-1, -2, 3) | Got: (-1, 0, 1) | ❌ FAIL
Trial 15/20 | Expected: (4, -4, 0) | Got: (2, -3, 0) | ❌ FAIL
Trial 16/20 | Expected: (-1, -1, 0) | Got: (1, 0, 0) | ❌ FAIL
Trial 17/20 | Expected: (-1, 1, -2) | Got: (-1, -3, 2) | ❌ FAIL
Trial 18/20 | Expected: (-1, -2, -1) | Got: (0, -1, -2) | ❌ FAIL
Trial 19/20 | Expected: (-4, 1, -1) | Got: (-4, 0, 1) | ❌ FAIL
Trial 20/20 | Expected: (1, 1, 0) | Got: (2, -1, -2) | ❌ FAIL
➡️ Depth 10 Completed. Average Success: 5.0%

--- Running Tests for Depth: 15 ---
Trial 01/20 | Expected: (-3, 4, 2) | Got: (-2, -2, 1) | ❌ FAIL
Trial 02/20 | Expected: (1, 4, 0) | Got: (1, 3, 1) | ❌ FAIL
Trial 03/20 | Expected: (2, 4, 3) | Got: (1, -1, 4) | ❌ FAIL
Trial 04/20 | Expected: (4, 2, 1) | Got: (4, 2, 1) | ✅ PASS
Trial 05/20 | Expected: (-2, 0, -1) | Got: (-2, 0, -2) | ❌ FAIL
Trial 06/20 | Expected: (-1, 2, 2) | Got: (-1, 0, 4) | ❌ FAIL
Trial 07/20 | Expected: (-3, 3, 1) | Got: (-6, 0, 0) | ❌ FAIL
Trial 08/20 | Expected: (1, -3, 5) | Got: (1, 1, 1) | ❌ FAIL
Trial 09/20 | Expected: (-4, -2, -1) | Got: (-4, -3, 0) | ❌ FAIL
Trial 10/20 | Expected: (1, 4, 2) | Got: (2, 2, 3) | ❌ FAIL
Trial 11/20 | Expected: (-5, 2, 0) | Got: (-5, -2, 4) | ❌ FAIL
Trial 12/20 | Expected: (-3, 2, -4) | Got: (-3, 1, -2) | ❌ FAIL
Trial 13/20 | Expected: (-1, -1, -1) | Got: (1, -2, 0) | ❌ FAIL
Trial 14/20 | Expected: (-2, 1, 0) | Got: (-2, -1, 2) | ❌ FAIL
Trial 15/20 | Expected: (3, -2, -2) | Got: (2, -1, -1) | ❌ FAIL
Trial 16/20 | Expected: (-6, 0, 1) | Got: (-6, 0, 2) | ❌ FAIL
Trial 17/20 | Expected: (-3, -1, -3) | Got: (-3, -4, 0) | ❌ FAIL
Trial 18/20 | Expected: (0, -2, 1) | Got: (-1, -1, 2) | ❌ FAIL
Trial 19/20 | Expected: (0, 0, 1) | Got: (-1, 0, 2) | ❌ FAIL
Trial 20/20 | Expected: (-1, 0, 2) | Got: (-1, -1, 3) | ❌ FAIL
➡️ Depth 15 Completed. Average Success: 5.0%

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🏆 FINAL COGNITIVE EVENT HORIZON REPORT | 3D MODE
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Graph Depth	Avg Context Tokens	Success Rate
2	51	60.0%
4	75	40.0%
5	87	40.0%
7	112	10.0%
10	149	5.0%
15	210	5.0%

==========

The data reveals a clear and definitive pattern. The failure mode of the Gemma 1B model is completely isolated to **Cognitive Load (Attention Head Saturation)**, while its Context Length remains perfectly stable.

Here is the step-by-step scientific deconstruction of the test results and the exact engineering limits that must be applied to the evolutionary graph.

1. Variable Isolation

- **1D Mode (Context Window Test):** At Depth 15, tracking a single variable over ~200 tokens, the model maintains 70% accuracy. This proves the model does not "forget" earlier tokens due to sequence length.
- **2D & 3D Modes (Cognitive Load Test):** The moment independent variables are introduced, performance collapses. In 3D mode, accuracy drops to 60% immediately at Depth 2, stabilizes briefly at 40% for Depths 4 and 5, and then catastrophically fails at Depth 7 (10%).

2. Mapping to CLUTTR

The CLUTTR dataset is an N-dimensional state-tracking problem. To resolve family trees, the model must hold multiple distinct entities in active memory (Marie, John, Anne, Lucie) and update complex, non-linear semantic vectors between them (brother, stepsister, mother).

Computationally, tracking semantic kinship vectors requires significantly more parameter activation than tracking simple arithmetic integers (+1, -1) across an X, Y, Z axis. Therefore, the **3D Mode** is the absolute minimum baseline for CLUTTR's complexity, and CLUTTR performance will degrade slightly faster than the 3D curve.

3. Calculating the Node Limit

The primary objective is to break the 20% baseline accuracy and push toward 30%+.

Looking at the 3D test curve:

- **Depth 2:** 60% Accuracy
- **Depth 4:** 40% Accuracy
- **Depth 5:** 40% Accuracy
- **Depth 7:** 10% Accuracy (The Event Horizon)

To achieve a 30% success rate on CLUTTR, the execution path must remain in a cognitive zone where the model's base accuracy is mathematically higher than 30%. In the 3D test, this zone ends abruptly after Depth 5. Once a graph reaches 7 reasoning steps, the attention mechanism fractures, yielding a 10% success rate (worse than the current CLUTTR baseline).

The Engineering Verdict: Maximum 4 Nodes

The Speciation Engine must enforce a strict **Maximum Path Depth of 4 Nodes** (with an absolute hard cap at 5).